

Mildew Maverick: UV-C Robotic Rover for Chemical-Free Grape Powdery Mildew Control with AI-Based Disease Detection

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Abstract

In this paper, we introduce Mildew Maverick, a UV-C light based robotic rover for powdery mildew control in traditional mandap/bower grape cultivation systems in India without using any chemicals. The rover is controlled remotely using the Blynk IoT mobile app and has an ESP32-WROOM-32E microcontroller as its main controller. The combination of an MLX90614 GY-9061 infrared thermometer for human safety detection, an HC-SR04 ultrasonic sensor for real-time obstacle avoidance and a WS2812B addressable LED strip, which is used to simulate the UV-C irradiation at different intensities according to the severity of mildew. An ESP32-CAM module with an OV2640 camera is used to detect grape diseases using AI with Roboflow API, using a 2-bit signal encoding scheme to send severity data to the main controller. The system provides an independent row-end navigation mode with automatic U-turn function, data logging to Google Sheets (by Google Apps Script) session by session, and generation of detailed session reports, such as number of rows treated, number of detections, severity distribution, and risk score. The studies by Gadoury et al. (2023) and McDaniel et al. (2024) have concluded that UV-C treatment is effective at removing 95% of grapevine powdery mildew without any residues of the treatment. Mildew Maverick connects the dots of this research into an easily accessible, low cost, IoT based agricultural robotics solution for Indian viticulture.

This could be an excellent solution for chemical-free viticulture and detecting grape diseases, such as powdery mildew. It would also be a great solution for the detection of grape diseases like powdery mildew, chemical-free viticulture, and autonomous navigation solutions that rely on an ESP32 microcontroller, the Blynk IoT cloud platform, and Roboflow.

I. INTRODUCTION

Erysiphe necator is the causal agent of powdery mildew of vineyards and is one of the most economically important diseases of vineyards worldwide and is more prevalent in the humid microclimate created by the traditional mandap/bower culture of vineyards in India. Powdery mildew is a common disease of table grape found in Maharashtra (India) and may cause a loss of 30 to 50% yield in a single season, if not controlled. Regular application of systemic fungicides like sulphur, myclobutanil and trifloxystrobin are used in conventional control. Although it is effective, this strategy can lead to pesticide residue on exported grapes, environmental damage and fungicide resistance in pathogen populations.

UV-C radiation (wavelength 254 nm) has been recently proven to be a scientifically proven alternative to control mildew. Gadoury et al. (2023) were able to reduce conidial density of E. necator on the surface of grapevine using UV-C treatment between 0.5 and 1.0 kJ/m² at night without causing any phototoxic damage. McDaniel et al. (2024) also demonstrated dose-dependent efficacy from a number of vineyard trials. However, while all this evidence is positive, the implementation has only been seen on larger scale mechanised in the developed countries, and this has not been a practical and affordable option for smallholder farmers in developing countries.

Mildew Maverick fills this void by combining UV-C LED simulation, AI-based disease detection, IoT-based remote control and cloud data analytics into a low-cost robotic platform that is specifically designed for the unique canopy geometry of the mandap/bower system. Controlled directly by the farmer with the use of a smartphone application, the rover can be operated without any robotic skills, though with optional autonomous row navigation. Session-to-session comparisons can be made using real-time disease data recorded on cloud storage, which can help with data-driven treatment decisions.

The system architecture, the hardware integration, the software design, the autonomous navigation strategy and the cloud data pipeline are described in this paper, as well as the results of the preliminary functional testing. A novel 2-bit severity signal encoding between ESP32-CAM and main ESP32 for real-time modulation of the UV intensity, an autonomous row-end U-turn algorithm based on HC-SR04 proximity sensor, a session reporting and multi-session comparison framework in Google Sheets and a human safety mechanism using passive infrared thermometry which cause immediate UV shut-off and motor stop if body-temperature objects are detected within the rover operating radius.

II. METHODS AND MATERIALS

A. Components Used

The hardware platform of Mildew Maverick integrates the following components:

Component	Model / Specification	Quantity	Role in System
Microcontroller	ESP32-WROOM-32E (240 MHz, WiFi + BT)	1	Main brain — runs all control logic
Camera Module	ESP32-CAM AI-Thinker OV2640	1	Disease detection, live stream
Motor Driver	TB6612FNG Dual H-Bridge	2	Drive 4 BO motors
IR Thermometer	MLX90614 GY-906 BCC (I2C, 3.3V)	1	Human safety detection
Ultrasonic Sensor	HC-SR04 (5V, 2 cm - 400 cm)	1	Obstacle + row-end detection
Logic Level Converter	Bidirectional 5V / 3.3V LLC	1	Protect ESP32 ECHO pin
LED Strip	WS2812B 12-LED Addressable RGB	1 strip	UV-C simulation, status display
DC Gear Motors	BO Motor 300 RPM Dual Shaft	4	4WD chassis drive
Power Supply	USB Power Bank 5V / 2A min.	1	Primary rover power
Buffer Capacitor	1000 uF / 16V Electrolytic	1	Prevent power bank shutoff
Control App	Blynk IoT (iOS / Android)	—	Remote control + telemetry

Cloud Storage	Google Sheets + Apps Script	—	Session data logging
AI Inference	Roboflow Universe API	—	Mildew classification

B. System Architecture

The system is organised into four functional layers: (1) actuation, (2) sensing, (3) control and computation, and (4) cloud analytics. The actuation layer consists of four BO gear motors powered by two TB6612FNG H-bridge modules and the WS2812B addressable LED strip. The sensing layer consists of an HC-SR04 ultrasonic distance sensor to detect obstacles and row end, an MLX90614 infrared thermometer for passive human detection and an OV2640 camera for disease detection using image information. This is done using the ESP32-WROOM-32E, which connects to the Blynk cloud via Wi-Fi and handles all timed tasks through the Blynk Timer library. The ESP32-CAM uses a 2-bit GPIO signal to send severity data to the main ESP32 as an independent co-processor. In the cloud analytics layer, Google Apps Script serves as an HTTP endpoint to accept structured JSON data payloads and populate Google Sheets with detection events and session summaries.

Data and control flow between the components of the system are shown in the following block diagram:

Layer	Component / Module	Interface	Direction
Cloud	Google Sheets (Sessions + Detections + Dashboard)	HTTPS / Apps Script	ESP32 OUT
Cloud	Roboflow Inference API (mildew detection)	HTTPS REST	CAM OUT
Control	ESP32-WROOM-32E (Main Controller)	Central Hub	—
Control	ESP32-CAM (Disease Co-processor)	GPIO 2-bit + WiFi	Bidirectional
Control	Blynk IoT Mobile App (V0–V15)	WiFi / Blynk Cloud	Bidirectional
Sensing	MLX90614 GY-906 (Human Safety)	I2C (GPIO 21/22)	IN to ESP32
Sensing	HC-SR04 + LLC (Obstacle / Row-End)	GPIO 14 / 12	IN to ESP32
Actuation	TB6612FNG x2 + BO Motors x4 (4WD)	GPIO 25/26/27/32/33/13	OUT from ESP32
Actuation	WS2812B LED Strip x12 (UV-C Sim.)	GPIO 15 (single wire)	OUT from ESP32

C. Methodology

1.Remote Control via Blynk IoT:-The rover is controlled using a custom Blynk dashboard with 16 virtual data streams (V0–V15) from a distance. The forward/backward and turn commands from a joystick widget mapped to V0 (Y-axis) and V1 (X-axis) are sent at 20

Hz. The motor speeds are calculated based on the formula: $\text{Left} = (Y + X) * \text{speed_scale}$ and $\text{Right} = (Y - X) * \text{speed_scale}$; speed_scale is a farmer configurable slider on V10. Temperature, distance, UV Intensity percentage, mildew severity level, row number and session elapsed time is pushed real-time to the app at 2.5 Hz with BlynkTimer tasks.

2.Human Safety Detection: – The MLX90614 GY-906 infrared thermometer is connected to the ESP32's hardware I2C bus (GPIO 21: SDA, GPIO 22: SCL), and reads the temperature of an object-surface at 5Hz. Objects with human body temperature can be detected within 33.5°C to 39.5°C, and the system automatically stops all four motors and flashes red LEDs of WS2812B for 100ms upon detection — the highest priority response in the system, and the response which overrides all other responses including independent navigation. The safety lock remains active until the thermal signature moves out of the detection window, so the UV-C LEDs are not able to irradiate personnel.

3.Obstacle Detection and Autonomous Row-End: - Navigation uses the HC-SR04 ultrasonic sensor, which is connected to ESP32 GPIO 12 using a Logic Level Converter (Bi-Directional) to protect the device's GPIO pins. The sensor operates at 5 Hz. The system sets two thresholds: the obstacle warning zone is at 40cm (by the orange LED) and the hard stop zone is at 22cm (when the motor stops if moving forward). The system considers reading < 18cm as row-end for autonomous row navigation (this is activated by Blynk V11). The U-turn sequence is a blocking state machine, meaning that the movement occurs in the following sequence: (1) Back 600ms (155 PWM), (2) Turn 90° right (480ms), (3) Sideways 1 rover (700ms) (4) Turn 90° right (480ms), (5) Forward again. The row count is automatically incremented and pushed to Blynk V12. Timing constants TURN_90_MS and SIDE_STEP_MS have been calibrated to the particular floor surface to which the deployment is taking place.

4.AI Disease Detection and UV Intensity Modulation: - The ESP32-CAM will operate a separate firmware loop that takes photos of JPEG frames every 8 seconds and sends the base64-encoded images to the Roboflow Inference API. The ArduinoJson library is used to interpret the API response for the prediction confidence and class labels. Any detections that match classes associated with mildew (powdery, mildew, fungal, disease and infected) and have a confidence score greater than 45% are assigned to one of three severity categories: LOW (one detection with a confidence score of 45–65%), MEDIUM (two detections or a confidence score of 65–85%) and HIGH (three detections or a confidence score of >85%). The severity is programmed in a 2-bit field on GPIO 12 (BIT0) and GPIO 13 (BIT1) of the CAM module, which are connected to GPIO 4 and GPIO 5 of the main ESP32 module, respectively. These pins are read by the main controller, which correlates the severity to brightness of WS2812B: 55 (None), 110 (Low), 190 (Medium), 255 (High). When the severity level is set to 3 (HIGH), LEDs turn to full white output to optimize the energy output of UV-C energy to the canopy.

5.Session Tracking and Cloud Data Pipeline: - When the farmer switches to autonomous mode or manually starts using Blynk, a session will be initiated with the data stored in the cloud. The recorded main ESP32 tracks are: rows completed, total mildew detections, severity distribution (high, medium, low hit counts), maximum severity level encountered, maximum ambient temperature and session elapsed time throughout the session. At the end of the session (when the Blynk V14 button is pressed), a structured HTTP POST request is sent to the endpoint of a Google Apps Script Web App with a JSON payload. The Apps Script automatically fills in three sheets: 'Detections' (a log table per event from the ESP32-

CAM), 'Sessions' (a summary table per session, with a computed Risk Score: HIGH x 3 + MEDIUM x 2 + LOW x 1), and 'Dashboard' (cross-session statistics, trend analysis, and colour-coding of risk-levels). The Blynk V15 display widget also receives session reports in real-time.

III. RESULTS

The Mildew Maverick system was subject to functional testing in the following seven evaluation areas. The rover was operated using Wi-Fi and a power bank and tested on a flat surface indoors that simulates a compacted-soil vineyard floor.

Test Area	Metric Evaluated	Result
Blynk Remote Control	Joystick latency, motor response	All 4 motors respond within 80 ms of joystick input; speed slider correctly scales PWM duty cycle
Human Safety System	Detection range, response time	MLX90614 detects hand at 35-38 C within 40 cm; motors stop and LEDs flash red within 1 sensor cycle (200 ms)
Obstacle Detection	Distance accuracy, stop behaviour	HC-SR04 reads within +/- 2 cm of measured distance; rover stops when obstacle enters 22 cm zone during forward travel
Autonomous Row-End Turn	U-turn completion, row count	Four-step U-turn sequence completes reliably; row counter increments correctly; TURN_90_MS calibrated to 480 ms on tile floor
UV Intensity Modulation	Brightness vs. severity level	WS2812B brightness correctly set to 55/110/190/255 corresponding to severity 0/1/2/3; level 3 triggers full-white output
ESP32-CAM Stream	Video stream availability	MJPEG stream accessible at http://[CAM_IP]:81/stream ; ~10 FPS at VGA resolution with PSRAM
Session Report + Cloud Log	Data accuracy, upload success	Session JSON uploaded via Apps Script HTTP 200; all fields (rows, detections, severity breakdown, risk score) correctly populated in Google Sheets

WS2812B boot animation (sequential red, green and purple sweep) worked as expected on power on, to give visual indication of LED strip connections. The Blynk Session Timer widget timed the session pretty precisely in MM:SS format. The Google Sheets Dashboard sheet was able to accurately calculate the cross-session risk score trends and classify disease

progression as WORSENING, IMPROVING or STABLE using a 10% threshold comparison between subsequent sessions.

IV. DISCUSSION

Mildew Maverick proves that UV-C robotics can not only be a high capital investment for big agricultural enterprises, but it can also be a cost-effective solution for smaller farms. The system also includes AI disease detection, safety mechanisms, autonomous navigation and cloud analytics, which are normally found in much more expensive precision agriculture platforms, while benefiting from building on commercially available microcontroller platforms and cloud-based APIs, resulting in a sub-US\$15 bill-of-materials (at Indian retail price).

The 2-bit signal encoding between the ESP32-CAM and main ESP32 is a design decision over using UART or I2C communication. UART would need careful baud rate synchronisation and would be prone to framing errors in noisy environments, while I2C would introduce addressing complexity for an additional microcontroller. The 4-bit GPIO solution provides 4 severities and close to zero latency with the clock domains of both processors fully isolated electrically.

The human safety detection system, based on the MLX90614, is a practical approach to ensuring that people are protected, in an agricultural environment, while working in proximity to UV-C. Passive infrared person detection is different from camera-based detection in that it is inherently real-time, without the need for heavy computation for inference, and it is also power-efficient. This detection range of 33.5°C to 39.5°C has been selected to minimize false positives from warm, non-human objects (e.g., sun-heated grape posts) while ensuring detection of skin surface temperatures within 50 cm. False negatives are reduced by the sensor's 90° field of view, and by the slow speed at which the rover travels in autonomous mode.

The Google Sheets multi-session comparison framework converts raw sensor data to vineyard intelligence. The Risk Score is a single quantitative metric (HIGH x 3, MEDIUM x 2, LOW x 1) which a farmer can track from week-to-week without needing to have any data science skills. With a risk score that is steadily decreasing, after each of the UV-C treatment sessions, farmers can empirically assess the effectiveness of the treatment and determine the frequency of treatment. This is part of an overall trend of digital farming records that are increasingly needed to meet export quality requirements.

There are several drawbacks to the present prototype that should be noted. The WS2812B LED strip is made with broadband visible and near UV light, not the germicidal UV-C wavelengths of 254 nm; germicidal UV-C LEDs with this wavelength are much more costly and must be housed in safety rated optical enclosures. The purple LED output is used for the hackathon demonstration as a useful approximation of the intensity of the delivery of UV-C. The strip would be replaced in reality by calibrated UV-C LEDs. Further, the detection using Roboflow is based on a public grape disease model whose performance on Indian grape varieties and the specific lighting conditions found in mandap systems has not been evaluated independently.

V. CONCLUSION

Mildew Maverick offers a full solution to the problem of chemical-free powdery mildew management in traditional bower grape farming for India with an affordable price tag and an added advantage of being an IoT based agricultural robotics solution. It successfully

combines variable-intensity UV-C light delivery, AI-based disease detection, infrared human safety protection, autonomous row navigation with multi-session cloud data analytics on a 4WD platform controlled by a smartphone.

The project highlights the potential for smallholder farmers to gain access to precision agriculture tools, typically associated with large agribusinesses, using appropriate open-source microcontroller ecosystems, free-tier inference APIs for AI, and cloud-based spreadsheet tools. The multi-session risk score tracking framework is especially a step towards data-driven viticulture practices in India.

Future work will include: replacement of the WS2812B strip with calibrated 265 nm UV-C LEDs with appropriate safety interlocks; training a custom Roboflow model under the specific light conditions of mandap and integration with a GPS module (Neo-6M) to present geospatially referenced disease maps; development of a native mobile application to integrate the Blynk control interface with the Google Sheets analytics dashboard and; formal field trial in a commercial vineyard to validate UV-C dose delivery with mildew incidence measurements.

VI. REFERENCES

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Conflict of Interest Statement:

The authors declare that there is no conflict of interest regarding the publication of this paper.

Acknowledgments:

The authors gratefully acknowledge the guidance and support of Prof. Shital Mali throughout the design, development, and documentation of this project. The team thanks the Department of Electronics and Communication Engineering, MIT-ADT University, Loni Kalbhore, Pune, for providing laboratory infrastructure and access to hardware components. The authors also acknowledge the open-source communities behind Espressif ESP32, the Blynk IoT platform, the Roboflow Universe public model repository, and the Adafruit NeoPixel library, whose freely available tools made this work possible.